

Investigating synthetic medical time-series resemblance

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ABSTRACT

Access to private medical data is restricted due to privacy laws, hindering research and real-world use. Synthetic data generation provides a viable solution by generating data with high utility and privacy protection without releasing the real data. Healthcare data records are often longitudinal in nature, being affected by covariates like age, gender, ethnicity, etc. As a result, synthetic healthcare data generation falls in the domain of time-series modeling and requires time-series based measures to investigate real and synthetic data resemblance. Covariate plots can be used for qualitative time-series resemblance but lack an empirical quantitative measure, thus, resulting in interpretations biased towards viewer's perspective. In this paper, we describe four time-series metrics to quantitatively evaluate the real and synthetic time-series resemblance on datasets from previously published healthcare research studies, both public and private. We apply the metrics on covariate plots for synthetic datasets to investigate the resemblance and compare the results with baseline synthetic datasets. We infer that the metrics effectively capture the time-series resemblance between real and synthetic datasets. The results highlight varying degrees of resemblance across sub-groups of covariates and multivariate time-series.

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1. Introduction

Data based approaches that use Artificial Intelligence (AI), and Machine Learning (ML) are at the forefront of developing and aiding real-world solutions. These would be especially useful in the healthcare domain, since medical records are generated by millions of individuals daily. Access to healthcare data can enable the development of healthcare solutions such as using Electronic Medical Records (EMRs) for enhancing patient care [1], and using health data to streamline clinical research [2]. Further, healthcare data is also useful for risk analysis [3]. Using ML techniques in the public health domain has already been shown to be effective for identifying hospital readmission, outbreak detection and more [4].

However, even with the abundance of healthcare data and its bountiful benefits, research is limited due to restricted access to medical data. Privacy laws such as Health Insurance Portability and Accountability Act (HIPAA) in the United States [5,6] and General Data Protection Regulation (GDPR) in the European Union [7] restrict access to healthcare records to preserve patient sensitive

information. Although necessary, such restrictions hinder widespread access and prevent research reproducibility and development of better data-driven solutions. Alternatives to direct access, including data de-identification and binning, can result in information loss and thus, inadequate results. Furthermore, direct sharing of real data can lead to data leaks, which can be costly.

As a result, a better solution that can solve data restriction without compromising patient privacy is required. Synthetic healthcare data generated from the real data while ensuring high utility, resemblance and privacy protection is a promising solution. This would enable data collectors, managers and researchers to disseminate the synthetic version of the healthcare data along with their results, enabling widespread access and reproducibility. Synthetic data generation using combinations of data processing and deep learning methods has shown to be effective for heterogeneous healthcare data consisting of binary, categorical and continuous variables [8]. Such datasets are able to replicate real data statistics while ensuring privacy, working for both public and private dataset replication [8,9].

However, healthcare data records are often longitudinal in nature with each individual having multiple occurrences of medical events across their life. These events are often influenced by covariates such as age, gender, ethnicity, etc., and thus, synthetic data must also effectively capture the time-series trends in the real

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data covariates. Current research often compares the real and synthetic data using covariate plots which present trend lines for different groups of data. Yale et. al. [9] compared the average count of morbidities in Autistic children using such plots. Dash et al. [10] compared the average sleep times for the real and synthetic dataset across age groups to underscore the qualitative utility of the synthetic data which are shown in Fig. 1. Even though we see deviations between the real and synthetic plots, the severity is hard to capture without proper quantitative analysis. Thus, the interpretation of the effectiveness of synthetic data in capturing the real data trends is biased towards the viewer's perspective. As a result, there is a need for metrics that can quantify this resemblance.

In this paper, we extend the European Symposium on Artificial Neural Networks (ESANN) 2021 conference paper “Quantifying Resemblance of Synthetic Medical Time-Series” [11] to provide a comprehensive analysis of the real and synthetic data resemblance using time-series metrics and highlighting the importance for such an analysis in evaluating the synthetic data generated. In Section 2, we describe two datasets used in this study highlighting the sources, and relevant data selected for this work. In Section 3, we describe generation of four synthetic datasets using different methods, and define baseline datasets for comparison. The section describes four time-series metrics to quantify time-series resemblance in the synthetic data domain. In Section 4, we present the results for the metrics on the two real datasets, investigating the resemblance across sub-groups of covariates and multivariate time-series. Section 5 discusses the need for the metrics and the inferences drawn while Section 6 concludes the study.

2. Data

The study presents results from two datasets: American Time Use Survey (ATUS) dataset and Autism Spectrum Disorder (ASD) claims dataset.

2.1. American Time Use Survey (ATUS)

American Time Use Survey (ATUS) is a federal national survey that estimates how and where Americans spend their time doing various activities like volunteering, socializing, sleeping, etc. [12] While the dataset includes several events for each individual, we

restrict the dataset to consider only sleep times based on a previously published dataset and study [10].

The data set has over 30 K records. It includes the average time slept each hour for 30 h starting at 4:00 AM to 10:00 AM the next day. These 30 h form 30 time events for each record. The dataset also includes two covariates: gender (Male, and Female), and age (15–24, 25–34, 35–44, 45–54, 55–64, 65–74, and 75+). These records are appended by the day and month, resulting in a total of 34 features.

2.2. Autism Spectrum Disorder (ASD)

Autism Spectrum Disorder (ASD) claims data is a private dataset derived from the OptumLabs® Data Warehouse that includes the diagnosis of over 280 K individuals having different Comorbid Medical Conditions (CMCs). The dataset includes information across a 5-year enrollment period with 6-month time-steps for each CMC. The seven CMCs are: Auditory disorders (Auditory), Development delays (DD), Gastrointestinal symptoms (Gastro), Immune related conditions (Immune), Psychiatric disorders (Psychiatric), Seizure disorders (Seizure), and Sleep disorders (Sleep).

The dataset is a multivariate time-series with 10 time-steps for each CMC, forming a total of 70 features. These are appended by additional information about gender (Male, and Female), race (White, Hispanic, Black and Asian), region (South, Midwest, West, and Northeast) and a 3-cluster clustering done using K-Means. The clustering was performed on the real data and replicated on a synthetic data effectively [9], however, it is not used for our analysis as we are concerned with covariates and time-series features. For this dataset, we explore two covariates: race and gender. The dataset was accessed inside a secure private environment provided by OptumLabs. The basic analysis on the real data was conducted within the secure system, however, access to this real data is no longer available.

3. Materials and Methods

3.1. Synthetic Data Generation

We generate four synthetic datasets for each dataset from the real data, using three data generation methods. These methods

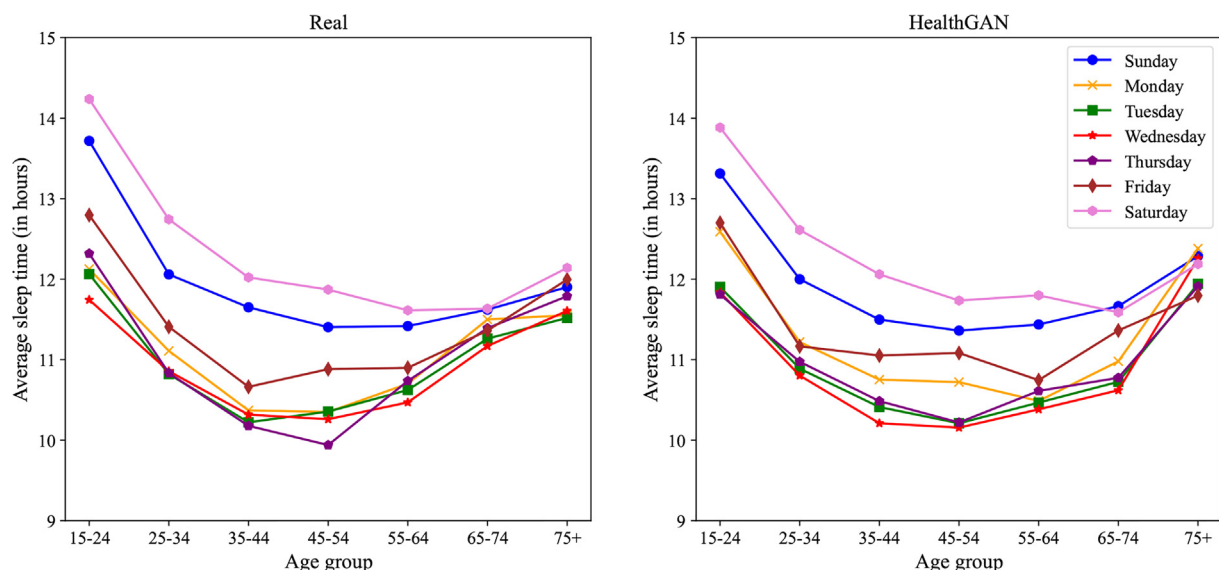


Fig. 1. Average sleep time in real and synthetic American Time Use Survey (ATUS) data [10]. Each trend line in the covariate plots describes the average sleep time for each day of the week, across several age groups in the data. Generally, the average sleep times are higher on weekends and for age groups 15–24 and 75+.

produce results with varying resemblance, often at the expense of privacy. However, they collectively provide a test-bed of datasets to compare and evaluate deviations introduced by synthetic data.

3.1.1. Bootstrapping

The first method is bootstrapping the synthetic data by randomly selecting rows from the real data with replacement, with the number of records equal to the training data. We refer to this synthetic dataset as *Bootstrap*. As the bootstrapped dataset includes actual real patient data, it will be very similar to the real data, resulting in high utility. However, this would lead to low privacy protection and such datasets could not be released for public use.

Although lacking privacy, this dataset forms an excellent baseline to understand how data generation could affect univariate and multivariate datasets, given that it preserves the real data statistics (mean, and variance) quite well. Further, comparison of other synthetic datasets with the baseline Bootstrap could reveal the variations introduced by other methods and at what cost.

3.1.2. Random Permutation

To introduce randomness in the synthetic data generated from the real data, we randomly permute a certain percentage of each column. The specified percentage of cells of each column are randomly selected and shuffled. We generate two datasets, at the two extremes, one with 10% random permutation of each column (referred to as *Rand10*) and the other with 90% random permutation of each column (referred to as *Rand90*). In both datasets, the univariate statistics would resemble the real data.

The *Rand10* dataset is likely to be very similar to the real and *Bootstrap* datasets, having similar utility but lower privacy protection. As a result, it is unlikely to be released for public use. On the other hand, *Rand90* is likely to suffer from significant loss in its resemblance to the real data as the inter-column dependencies and patterns must be severely impacted. For instance, stratification of the data based on the sub-groups of a covariate would lose the data trends as there is extreme randomness and the original distribution is likely to be affected. In this way, the two datasets form two baselines at the two extremes of the spectrum.

The datasets generated using *Rand10* and *Rand90* have been proposed for baseline comparison with synthetic data generators. While these datasets cannot be used for real-world applications due to their limited privacy preservation, they form baselines where *Rand10* is likely to have high resemblance with *Rand90* likely to lie on the other end of the spectrum.

3.1.3. HealthGAN

While synthetic data generation has been used across several applications such as in finance [13], image handling [14,15] etc., its application in healthcare specifically requires that it must preserve patient privacy. Generative Adversarial Networks (GANs) [16] based generators such as medGAN [17], HealthGAN [8], and DPGAN [18], provide one such solution to generating healthcare data with privacy protection, each presenting a different trade-off between utility, resemblance, and privacy.

HealthGAN has specifically shown to be better at protecting patient data while ensuring high utility and resemblance [17]. Furthermore, it has been found to be effective in even capturing time-series data for multiple healthcare datasets [10,19]. HealthGAN attempts to capture the multivariate data distribution in order to achieve high resemblance and utility, which might affect the covariate-level effectiveness. Based on the observed effectiveness of HealthGAN across a multitude of datasets, we use the data generated by it as the fourth synthetic dataset, and simply refer to it as *HealthGAN*.

3.2. Metrics for Time-Series Resemblance

Covariate plots include the time-series trends for the sub-groups of a covariate. These are used to compare the real and synthetic healthcare datasets, where covariate-level analysis is important. However, they lack a quantitative measure of this comparison. Thus, we use four time-series metrics to empirically evaluate the real and synthetic time-series covariate data resemblance. The metrics have been adapted from time-series clustering literature [20,21] to be used in the synthetic time-series domain. Let the real time-series be represented using $\{x^k\}_{k=1}^n$ while the corresponding synthetic time-series be represented using $\{y^k\}_{k=1}^n$, each series having the same time-steps n .

3.2.1. Root Mean Square Error (RMSE)

It evaluates the closeness between the real and synthetic data at every time-step to understand how much the synthetic data deviates from the real data on a per time-step basis. We normalize the values by dividing by the total sum of squares of the real data. The resultant mathematical formula is defined by Eq. 1. This normalization enables us to understand if the synthetic data is representing the real data better than if the synthetic data was simply producing the average of the real data (i.e. constant).

$$RMSE = \sqrt{\frac{\sum_{k=1}^n (x^k - y^k)^2}{\sum_{k=1}^n (x^k - \bar{x})^2}} \quad \bar{x} = \frac{\sum_{k=1}^n x^k}{n} \quad (1)$$

We then transform the RMSE values using Eq. 2 so that 1 indicates a perfect match while 0 indicates a complete mismatch. Due to the normalization introduced earlier, a value less than 0.5 would indicate that the synthetic data is performing worse than simply having an average value.

$$f(x) = \frac{1}{1+x} \quad (2)$$

3.2.2. Pearson's Correlation Coefficient (PEARSON)

It measures the real and synthetic data linear correlations, resulting in values between -1 to 1 . However, we transform the formula to define this correlation between 0 to 1 as seen in Eq. 3. In this modified formula, 1 implies a perfect correlation between the real and synthetic data while 0 indicates a perfect anti-correlation.

$$PEARSON = \frac{\frac{\text{cov}(x,y)}{\sigma_x \sigma_y} + 1}{2} \quad (3)$$

In Eq. 3, $\text{cov}()$ is the covariance between the real and synthetic data while σ indicates the standard deviation of the respective data.

3.2.3. Directional Symmetry (DS)

It measures the ability of synthetic data to capture the time-series direction of the real data (increasing or decreasing), from one point to the next. We divide the original mathematical formula by 100 , so that the values are captured between 0 and 1 . A value of 1 indicates that the direction of movement from one point to the next matched exactly between real and synthetic data while a value of 0 indicates that it did not match in any time-step.

$$DIRECTIONAL = \frac{1}{n-1} \sum_{k=1}^{n-1} d^k \quad (4)$$

Eq. 4 defines the mathematical formula for DS, where d^k takes a value of 1 if $(x^{k+1} - x^k)(y^{k+1} - y^k) \geq 0$ and 0 otherwise.

3.2.4. Short Time-Series Distance (STS)

It captures the difference between the relative values rather than the absolute values, where the time-series are considered as piece-wise linear [22]. We again normalize the values by dividing by the total sum of squares of the real data as shown in Eq. 5, where $\bar{x}$ is the mean of x^k . We transform the result to a value between 0 and 1 using previously defined transformation in Eq. 2. Note that, in our work, the difference $t^{k+1} - t^k$ is always 1.

$$STS = \frac{\sqrt{\sum_{k=1}^{n-1} (x^{k+1} - x^k)^2}}{\sqrt{\sum_{k=1}^{n-1} (x^k - \bar{x})^2}} \quad x^k = \frac{x^{k+1} - x^k}{t^{k+1} - t^k}, y^k = \frac{y^{k+1} - y^k}{t^{k+1} - t^k} \quad (5)$$

While other time-series metrics can also be used, they were not included due to redundancy. For example, RMSE is already a form of Euclidean distance and defining additional distance metrics based on Euclidean distance is not likely to add novel insights for resemblance. The metrics used in this paper emphasize that time-series metrics can be adapted to the synthetic time-series resemblance domain and the choice of which metrics to use is based on the application and dataset.

The four metrics in this paper provide a comprehensive analysis to quantify the time-series resemblance between the real and synthetic data. The metrics measure the resemblance with respect to properties we are interested in the healthcare domain: RMSE measures the total average discrepancy and is a common way of evaluating time series for regression, clustering and forecasting. PEARSON discounts for distortions due simply to shift and scale. DS and STS are two measures focusing on variations. While STS can be thought of as the RMSE for the series derivatives, DS focuses only on the concordance between upward and downward variations, hence is more forgiving. These enable us to dive deep and analyse the effect of generative models on producing high quality time-series healthcare data for real-world use.

To illustrate the difference between the metrics and highlight when they are effective in capturing the differences between real and synthetic datasets, we present two example scenarios. In each

scenario, we consider time-series datasets with 10 data points collected at time steps $t = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$ respectively.

In the first scenario, we have “Real” dataset defined by [105, 104, 103, 102, 101, 205, 204, 203, 202, 201]. We generate 5 synthetic datasets for comparison with this data: (a) “Real+80” which only shifts the real data by adding 80 to each value, (b) “Real $\times 1.6$ ” which only scales the real data by multiplying the values by 1.6, (c) “Real-mean” which creates a time-series with a constant value equal to the mean of the real data, (d) “Real-slope” which has the slope equal to mean slope of the real data, and (e) “Handmade” with values [101, 102, 103, 104, 105, 201, 202, 203, 204, 205] designed to illustrate the effect of slight variation from real data.

The set of synthetic datasets compared with “Real” along with the results of the four metrics are plot in Fig. 2. We see that neither shifting nor scaling effects PEARSON or DS, so they both perfectly match the real data in those metrics. We also see that shifting does not effect the STS. The “Real-mean” has an RMSE of exactly 0.5 due to our normalization of RMSE, PEARSON has undefined value since the synthetic series is constant, and DS is perfect due to the non-strict inequality in its definition. The “Real-slope” has STS value 0.5 due to our normalization. The “Handmade” has values close to the real data and similar slopes (most close to zero), hence RMSE, PEARSON, and STS are all high, but the DS is very poor.

In the second scenario, we have the real dataset “Real 2” with values equal to the first 10 perfect squares. We generate two synthetic datasets, “Handmade 2” and “Handmade 3”, that have overall increasing trend similar to the “Real 2” data with oscillating slopes to varying degrees. “Handmade 2” multiplies the “Real 2” values with [0.75, 1.2, 0.8, 1.15, 0.85, 1.1, 0.9, 1.05, 0.95, 1.0] while “Handmade 3” takes the values [-5, 10, 3, 23, 18, 44, 40, 73, 71, 105].

In Fig. 3, we can see the synthetic datasets deviating only moderately from the real data, but with alternating oscillations above and below. This leads to good RMSE values but poor STS results. With increased oscillation from the real data, the scores for all metrics except PEARSON get worse.

From the two examples, we show that the metrics are measuring different aspects of similarity between the real and synthetic time series and thus, should collectively be used for evaluating resemblance. Using them in conjunction ensures that various forms of deviations of the synthetic data from the real data are captured. For example, DS and STS measure for directional resemblance, while RMSE and PEARSON capture deviations in the values observed. As shown through the two examples above, some

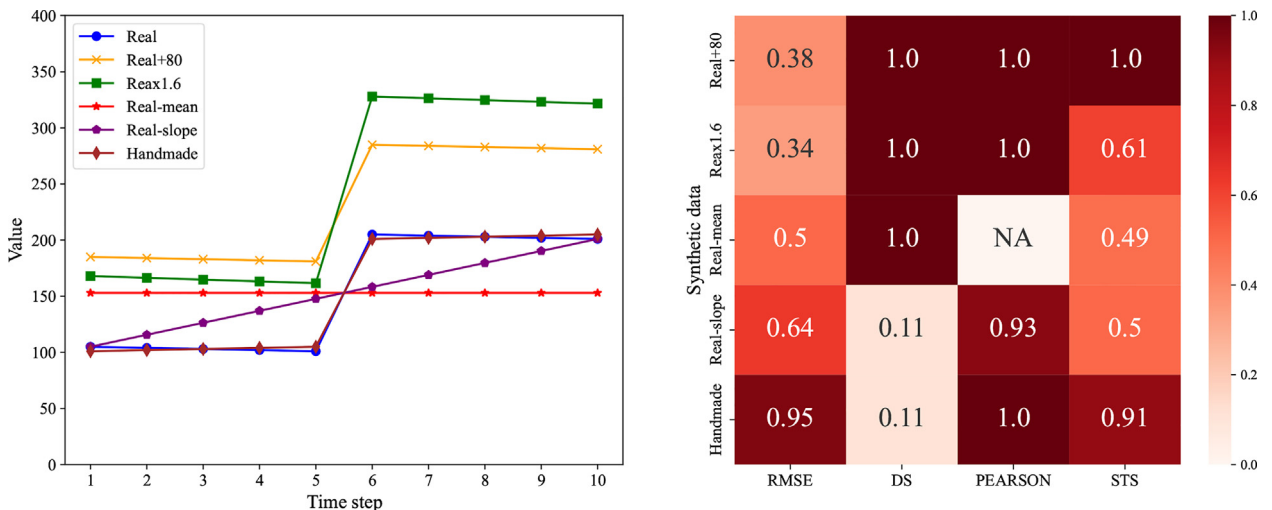


Fig. 2. Results of metric scores when comparing synthetic datasets to “Real” data. The plot on the left shows the datasets. The synthetic datasets: Real + 80, Realx1.6, Real-mean, Real-slope, and Handmade are compared with Real and the resulting effect on the four metrics is shown by the heatmap on the right. Values closer to 1 indicate good resemblance while those closer to 0, indicate bad resemblance.

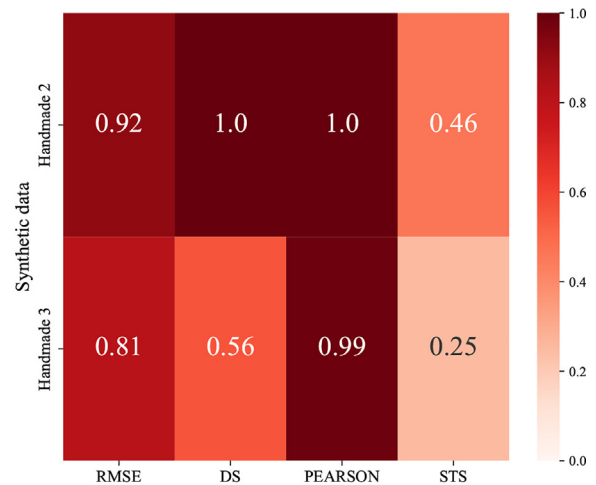
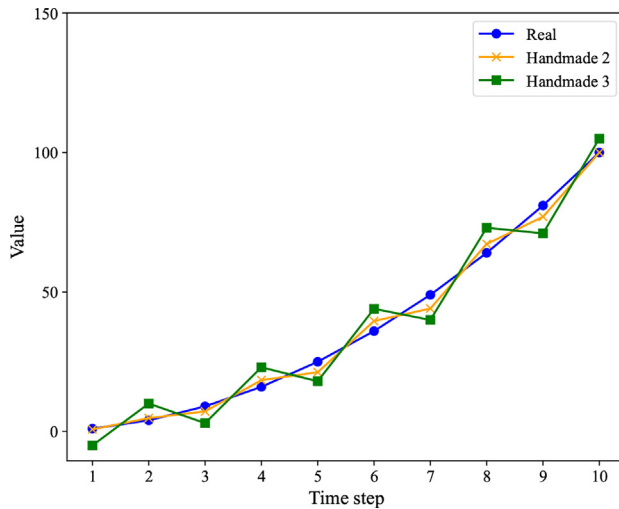


Fig. 3. Results of metric scores when comparing synthetic datasets to “Real 2” data. The plot on the left shows the datasets. The synthetic Handmade 2, and Handmade 3 are compared with Real 2 and the resulting effect on the four metrics is shown by the heatmap on the right. Values closer to 1 indicate good resemblance while those closer to 0, indicate bad resemblance.

deviations are captured only by certain metrics and not others. Thus, we recommend using all the proposed time-series metrics together as it provides a comprehensive look at the resemblance between the real and synthetic datasets. There are no specific thresholds for these metrics which serve as the demarcation between good and poor resemblance. Instead, these metrics should be used to compare the resemblance of two or more synthetic datasets to the real data. Using all of the metrics will ensure a more thorough comparison than using any one metric, and comparisons which use only one metric should be viewed with skepticism. The metrics and results are available via GitHub repository “SyntheticTimeSeriesMetrics” [23].

4. Experimental Results

We evaluate the four time-series metrics on two datasets as case studies to understand how well they work on univariate and multivariate healthcare datasets.

4.1. Case Study 1: ATUS

To create time-series trend plots for sub-groups of age, we group the data by age and then, day of the week. We then calculate the average sleep time in hours across the 30 time events for each such combination of age and day. We plot this change in average sleep time stratified by various age groups as time-series trend lines across the seven days of the week, starting on Sunday.

Fig. 4 shows the covariate plot for age in the real ATUS data. The real data results show that the trend across all age groups is very similar, having lower average sleep time during the weekdays but higher during the weekends. Age groups 15–24 and 75+, have the highest average sleep hours across majority of the days. Further, we find that the range of average sleep time varies across age groups. Age group 15–24 has the maximum range: 2.49 h while age group 65–74 has the least: 0.46 h. Ideally, the synthetic datasets should capture these trends while also preserving the variations.

Fig. 5 shows the covariate plots for synthetic ATUS datasets. For the synthetic results, we observe that both Bootstrap and Rand10 have very similar trend lines to each other and the real data. For HealthGAN, while the trend is very similar, there are some discrepancies. For example, both 15–24 and 75+ trend lines appear to be

different from the rest of the trend lines. The trends for other age groups also appear to be more sparse than the real data. Further, the Rand90 dataset completely loses its patterns and results in rather stagnant average sleep time across all age-groups. All age groups in this dataset appear to sleep for almost similar times which is completely different from the real data. However, even though some differences might be visually noticeable between the real and synthetic datasets, the actual impact needs to be quantified. This will endorse any interpretations we draw about the resemblance. Thus, we apply the four time-series metrics on these plots and compare them with the real data.

From Fig. 6, we observe that Bootstrap data performed the best across all metrics, being closely followed by the Rand10 dataset. Their values being less than 1 for RMSE, DS and STS show that even though the plots appear similar to the real data, the metrics show that there are still differences which are not apparent by visual inspection. The results also emphasize the effectiveness of the metrics in being able to capture the resemblance between real and synthetic trends well.

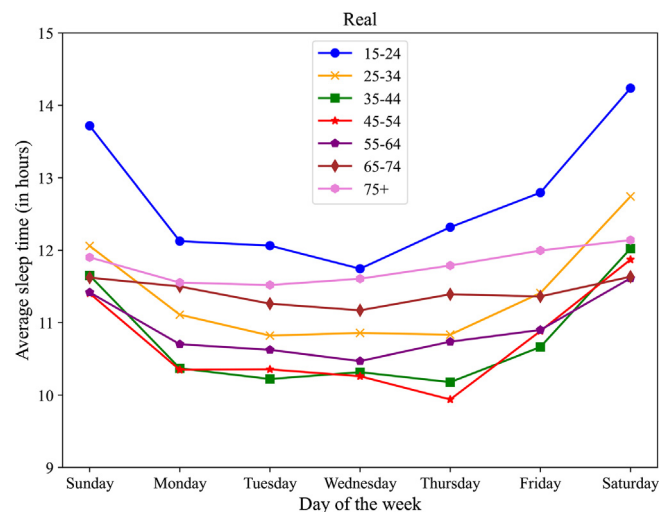


Fig. 4. Covariate plot of time-series trends for sub-groups of age in real ATUS dataset. The plot includes a trend line for each age group describing the average sleep time, in hours, slept across days of the week.

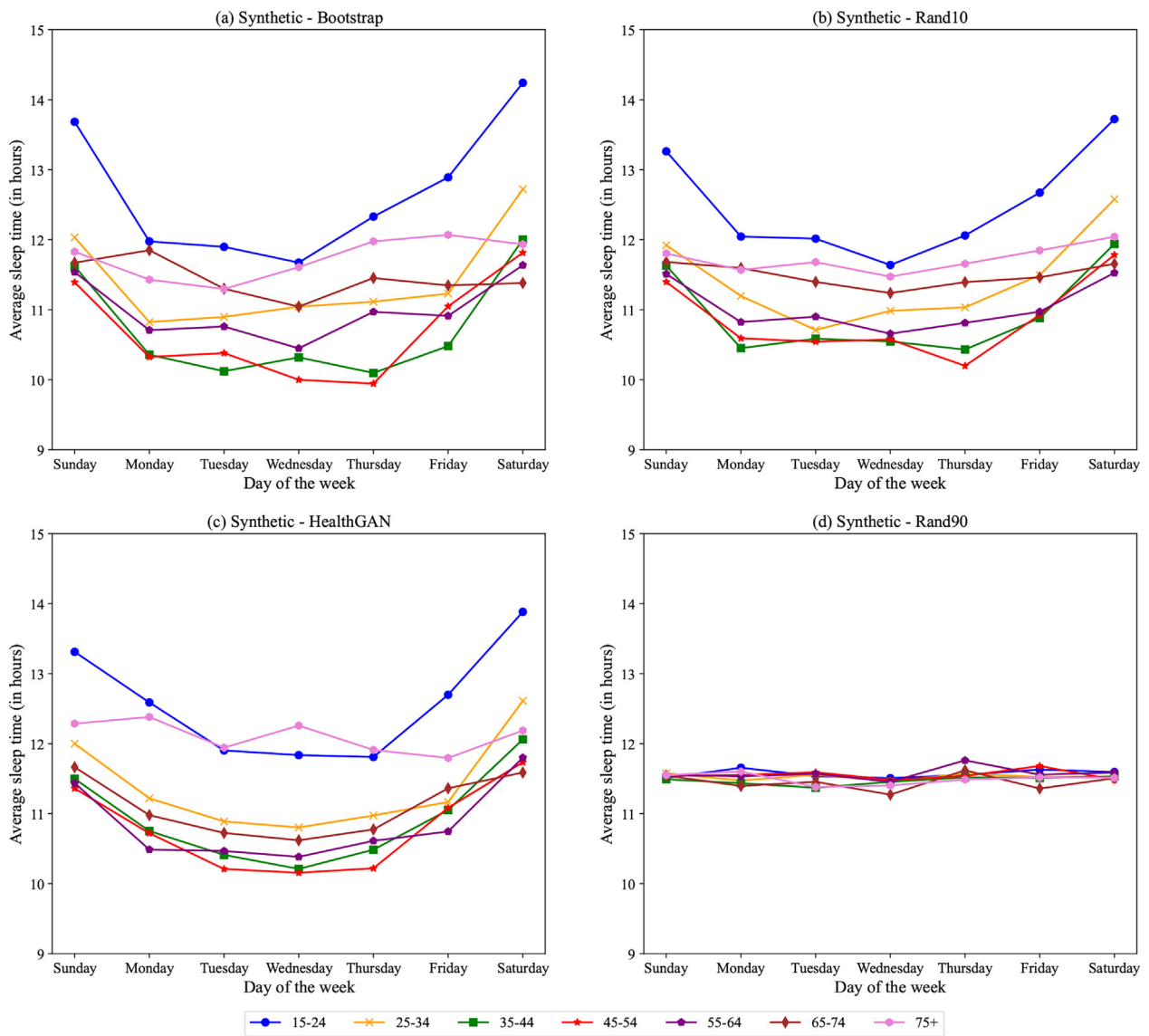


Fig. 5. Covariate plot of time-series trends for sub-groups of age in synthetic ATUS datasets. The plots includes trend line for each age group describing the average sleep time, in hours, slept across days of the week. The time-series trends are shown for all four synthetic datasets: (a) Bootstrap, (b) Rand10, (c) HealthGAN and (d) Rand90.

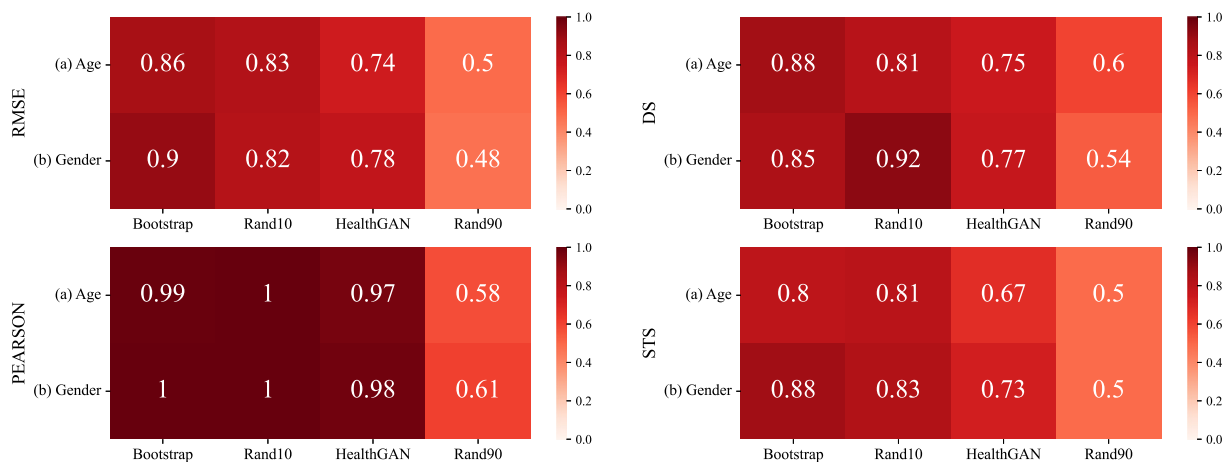


Fig. 6. Results of the four time-series metrics on covariates of: (a) age, and (b) gender, for the ATUS dataset. The results are presented for comparing the real data with four synthetic datasets: Bootstrap, Rand10, HealthGAN and Rand90. Values closer to 1 indicate good resemblance while those closer to 0, indicate bad resemblance.

We find that HealthGAN does a good job across all metrics, but the enforcement of privacy protection does come at a cost of resemblance. None the less, it provides the appropriate combination of privacy protection and resemblance for real-world application. The Rand90 data performs the worst, having values closer to 0.5, indicating it is equivalent to simply taking a random average. This dataset can not be used for any real application due to its lack of utility. However, for the ATUS data, this was expected as random permutation of 90% of the cells in each column is likely to disrupt the original data trends, and hence, bad performance.

Similar analysis is also performed on the sub-groups of the gender covariate in ATUS dataset, the results of which are presented in Fig. 6. The results are similar to our observations in the age covariate. The Bootstrap performs the best across 3 out of 4 metrics while 10% random permutation performed the best for DS. Hence, overall, Bootstrap was closely followed in results by Rand10. HealthGAN performed quite well, having resemblance scores only slightly lower than Bootstrap and Rand10. However, as it also ensures privacy protection, it is a great data for real-world distribution when measuring for real and synthetic covariate resemblance. Rand90 performed the worst amongst all synthetic datasets. We also note that the results are slightly better on gender than age, indicating that the synthetic data captured it more representatively.

However, evaluating the complete plot can mask the variations across different sub-groups of covariates. As the age saw relatively worse results than gender, we take a deep dive by measuring the resemblance across each age group individually to understand the resemblance for each sub-group. The results for three contrasting age groups: 15–24, 65–74, and 75+ are shown in Fig. 7.

When we observe the results across individual age groups, we find a noticeable deviation. The datasets have better scores for 15–24 age group than the other two age groups, indicating a higher resemblance. These differences are often exaggerated on the HealthGAN data. While in general Bootstrap and Rand10 performed the best, it is not true for all age groups. For example, metrics RMSE and STS show that Bootstrap performed worse than

Rand90 probably. Furthermore, HealthGAN consistently performed worse than even the Rand90 data for age groups 65–74 and 75+.

Application of these metrics highlighted the differences in resemblance of synthetic data without the need for closer visual inspection. The metrics captured the extent of the resemblance performance of the synthetic datasets. While the visual inspection showed deviation in some age groups, the actual differences were quantified by the metrics, surfacing age groups such as 65–74 which would have otherwise gone unnoticed. This strengthens the fact that combinations of these metrics can quantify even slight visual changes. From the results across age groups, we find that the synthetic datasets are not equitably representing the trends for various sub-groups. This suggests that the synthetic data as a replacement for real data might be effective for some sub-groups but not all.

4.2. Case Study 2: ASD

The ASD dataset is quite different from the ATUS dataset, having over 280 K records and being multivariate (7 CMCs). While the dataset includes the count of occurrence of each diagnosis across the time-steps, we transform the data to a binary form after synthetic data generation, based on whether the diagnosis occurred (1) or did not occur (0). The data is then transformed to evaluate the prevalence of each CMC across each time step creating a multivariate time-series of prevalence values and is used for this evaluation.

We generate covariate trend lines for each CMC and covariate, and compare the real and synthetic data combinations using the four time-series metrics. For our analysis, we look at two covariates: race and gender.

Fig. 8 shows the covariate plots for Auditory disorder in the race covariate of ASD synthetic datasets. Bootstrap and Rand10 look very similar to each other, and are likely to perform really well in resemblance. HealthGAN data looks similar for the first couple of years of the data but then starts to deviate. As seen during the last two years of the study, it increases and decreases randomly. Rand90 squishes the data lines together and has a consistent direc-

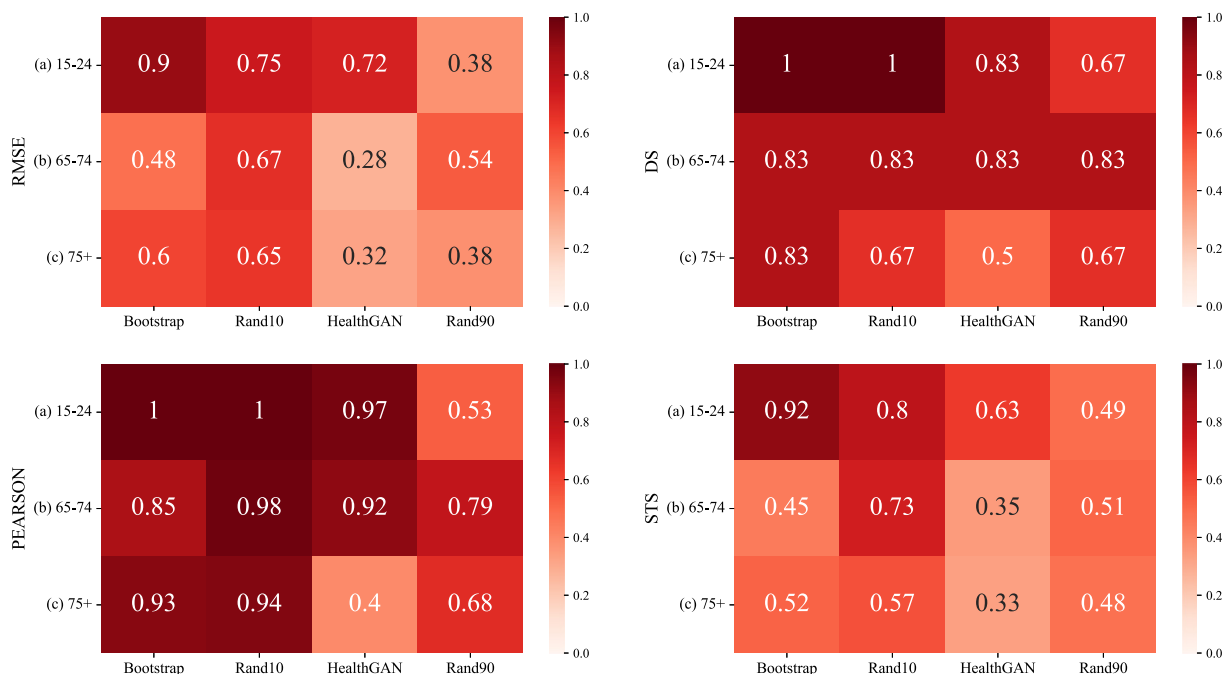


Fig. 7. Results of the four time-series metrics on three age-groups: (a) 15–24, (b) 65–74, and (c) 75+, for the ATUS dataset. The results are presented for comparing the real data with four synthetic datasets: Bootstrap, Rand10, HealthGAN and Rand90. Values closer to 1 indicate good resemblance while those closer to 0, indicate bad resemblance.

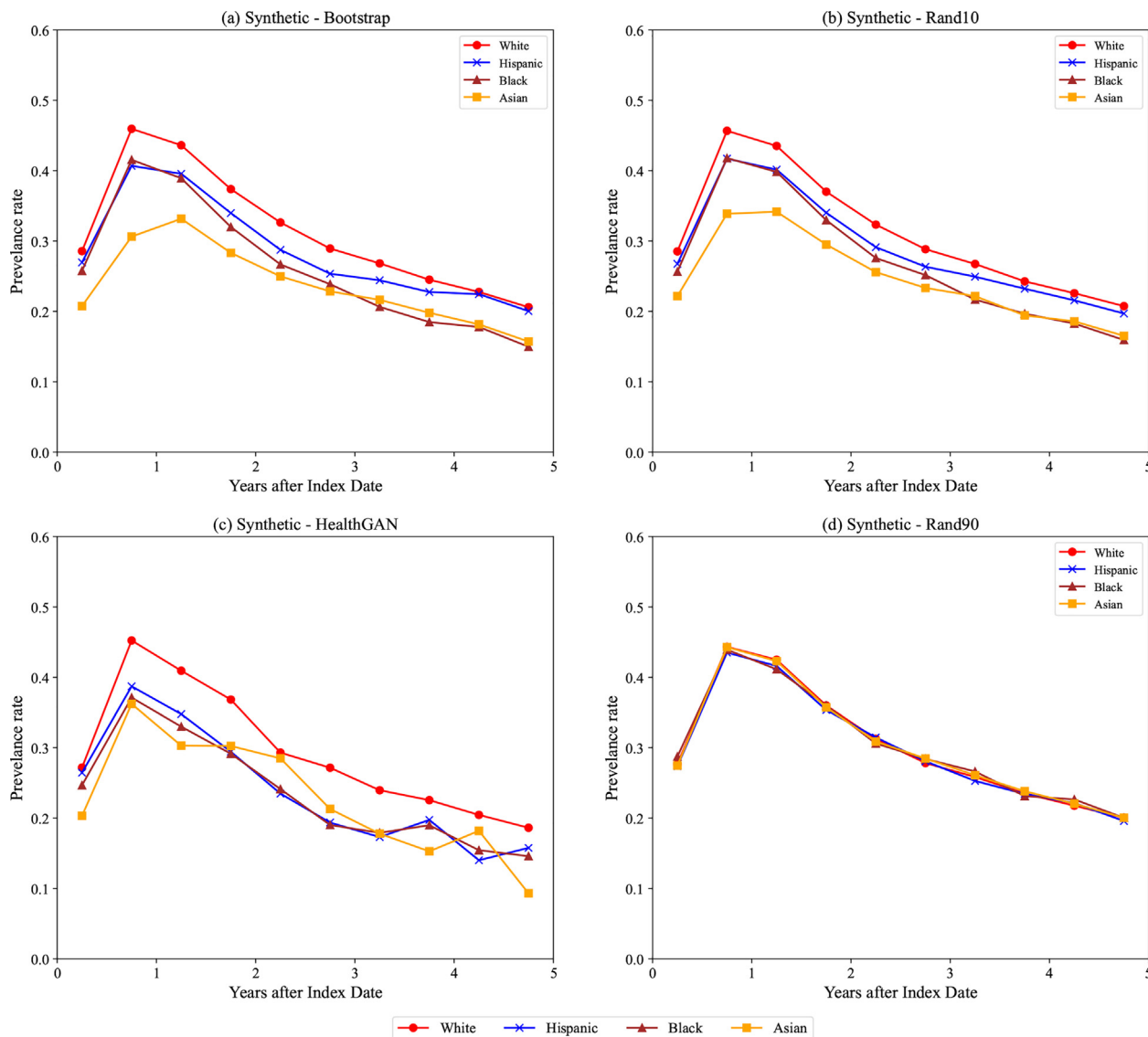


Fig. 8. Covariate plot of time-series trends for sub-groups of race in synthetic ASD datasets. The plots includes trend line for each race group describing the prevalence of the diagnosis across the 5 year enrollment period. The time-series trends are shown for all four synthetic datasets: (a) Bootstrap, (b) Rand10, (c) HealthGAN and (d) Rand90.

tion. This implies that it is likely to perform better on directional metrics than the non-directional metrics. To understand the empirical differences, we apply the time-series metrics on these plots. Similar plots and prevalence rates are calculated for the other 6 CMCs as well and the time-series resemblance metrics are evaluated. The results are all presented in Fig. 9.

From the results, we find that the Bootstrap and Rand10 data perform the best across all metrics. These are followed by results for Rand90, which is performing better than HealthGAN. The higher performance, even after extreme permutation, can be attributed to the fact that the values are sparse and binary. So, even permutation does not result in significant deviation from the original values. While directional metrics had higher resemblance scores as expected, RMSE does capture the lack of sparseness across sub groups of race in the Rand90 data when compared with the real data. The worst performing model in comparison is HealthGAN, indicating that it struggles to capture multiple time-series. This highlights that for capturing synthetic data trends while preserving privacy, there is a cost to resemblance. The results are worst for Seizure, especially for HealthGAN.

Table 1 shows the overall prevalence values for the Seizure disorder comparing the real and synthetic datasets. It becomes quite apparent from the table, why we saw a poor performance. While the prevalence rates in Seizure are already low, the synthetic data exaggerates these prevalence values across the time-series. For HealthGAN data, the prevalence rate increases by over 5x as seen for year 2.5 but is negligible for years 3 to 4. This is probably due to the very small prevalence rates of the real data from which HealthGAN data is synthetically derived. This, however, makes the HealthGAN synthetic data different from the real data. All other synthetic datasets have similar prevalence scores, and thus, perform better than HealthGAN.

We generate similar plots for the gender covariate in ASD data and evaluate the time-series metrics. The results are shown in Fig. 10. The results are quite similar to the race covariate. The best performing datasets are Bootstrap and Rand10, having similar scores to each other. The scores across CMCs were only slightly different with the worst performance for Seizure in STS. Rand90 had similar performance overall but performed worst for Immune and Seizure CMCs. However, HealthGAN performed the worst in comparison, especially for Seizure, with near 0 scores for RMSE and



Fig. 9. Results of the four time-series metrics for the 7 CMCs of the ASD dataset for race covariate. The results are presented for comparing the real data with four synthetic datasets: Bootstrap, Rand10, HealthGAN and Rand90. Values closer to 1 indicate good resemblance while those closer to 0, indicate bad resemblance.

Table 1

Prevalence rates for Seizure disorders as observed for the overall real and synthetic datasets across the 10 half-year periods.

Year	Real	Bootstrap	Rand10	HealthGAN	Rand90
0.5	0.0056	0.0058	0.0056	0.0171	0.0056
1.0	0.0061	0.0063	0.0061	0.0031	0.0061
1.5	0.0074	0.0074	0.0074	0.0086	0.0074
2.0	0.0064	0.0067	0.0064	0.0111	0.0064
2.5	0.0055	0.0055	0.0055	0.0245	0.0055
3.0	0.0049	0.0049	0.0049	0.0000	0.0049
3.5	0.0045	0.0045	0.0045	0.0000	0.0045
4.0	0.0042	0.0044	0.0042	0.0000	0.0042
4.5	0.0042	0.0045	0.0042	0.0038	0.0042
5.0	0.0041	0.0044	0.0041	0.0109	0.0041

STS. This is likely due to the HealthGAN struggling to capture the data with small prevalence scores well.

From the results, we observe that even though the models based on bootstrapping and random permutation form good baselines, they are less than ideal for real world use due to lack of privacy protection or usable data. On the other hand, HealthGAN provides a good combination of privacy protection and overall utility. However, it too struggles when the datasets are skewed and shows signs of bias towards certain sub-groups.

5. Discussion

Comparing real and synthetic datasets is very important for understanding if the synthetic data can act as a proxy for real data for healthcare applications. Using plots comparing the real and synthetic data trends is a common way of doing such a comparison. However, the inferences drawn from plots such as covariate plots are influenced by the domain expertise and knowledge of the reader and thus, do not provide a solution for quantitative comparison.

In this paper, we presented four time-series metrics that are adapted to the synthetic time-series domain for quantifying the resemblance between the real and synthetic datasets. While visual inspection of the covariate plots is not enough, the inferences strengthened by the results from the time-series metrics provide an optimal solution.

The metrics were evaluated on multiple covariates of two healthcare datasets including univariate and multivariate time-series. We found that it is essential to use multiple metrics simultaneously to capture several aspects of real and synthetic data resemblance including absolute values, relative values and directional matching. The collection of metrics ensures that all deviations of the synthetic data from the real data are appropriately captured. For example, in ASD dataset while the general trend was captured by Rand90 very well and so was confirmed by the directional metrics, RMSE underscored with a lower score that the absolute difference in values from the real data is significant.

The metrics also highlight resemblances or lack thereof between real and synthetic datasets which are difficult to identify through visual inspection. For example, identifying that certain age groups performed worse while others did not was evidently clear when observing the metrics on age groups individually for ATUS data. The results highlighted that age group 65–74 had sometimes worse performance than other groups.

Application of the various time-series metrics on the real and synthetic datasets revealed that the scores can vary across different sub groups and time-series. This suggests that even when the synthetic data resembles the real data overall, the results might differ at the individual sub group level. In Fig. 7 we saw that the metrics had higher scores for the age group 15–24 and lower scores for 65–74 and 75+. Furthermore, from Fig. 9, we found that the resemblance was much higher for Auditory disorder than Seizure disorder.

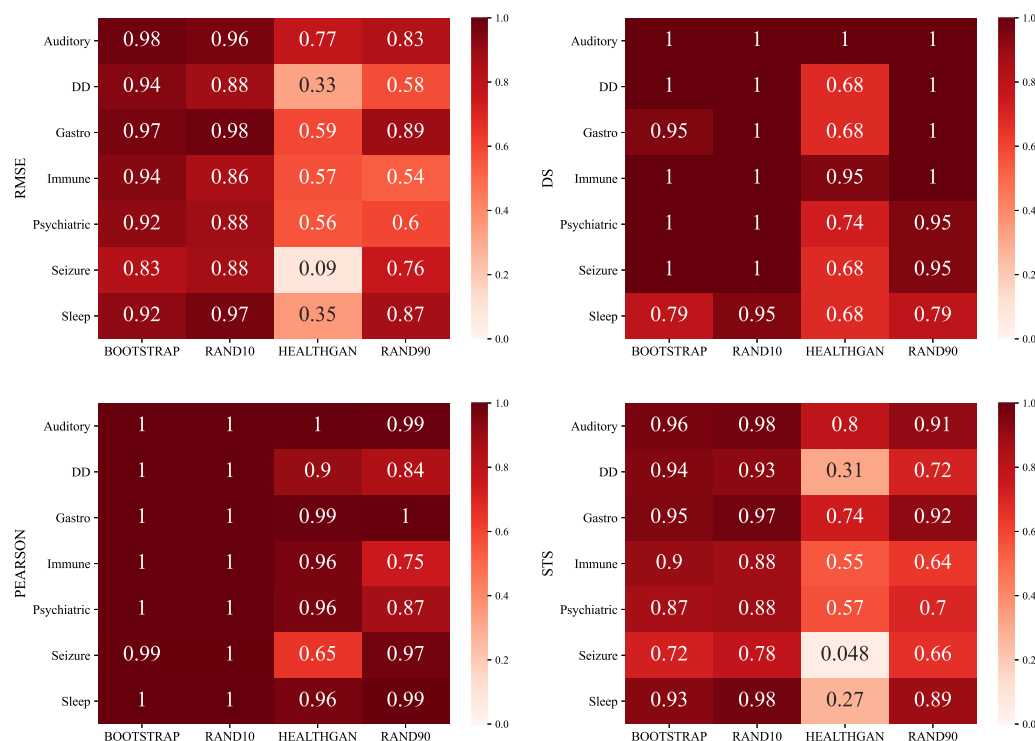


Fig. 10. Results of the four time-series metrics for the 7 CMCs of the ASD dataset for gender covariate. The results are presented for comparing the real data with four synthetic datasets: Bootstrap, Rand10, HealthGAN and Rand90. Values closer to 1 indicate good resemblance while those closer to 0, indicate bad resemblance.

The results suggest that synthetic data is not representing all groups and time-series as well, thus leading to biases. While real healthcare datasets have inequities, such differences in resemblance of synthetic datasets can exacerbate these inequities. As a result, if such synthetic data is used for creating real-world solutions, we are likely to observe models and solutions which might not work on the real data, leading to poor performance.

From the results, we infer that such a combination of time-series metrics is an essential quantification tool to understand and emphasize real and synthetic data resemblance. The metrics were useful in highlighting such discrepancies but require further exploration to understand if the problem of synthetic data bias is significant and how it might be mitigated.

6. Conclusion

Synthetic healthcare data is longitudinal and thus, synthetic data must capture the covariate time-series trends in the data. Thus, in this paper, we use four time-series metrics to compare the real and synthetic data resemblance and discuss the observed results. The four time-series metrics were collectively able to capture the various aspects of synthetic data resemblance. The metrics highlight resemblance of absolute and relative values along with directional match between real and synthetic datasets.

We found that synthetic baselines such as bootstrapping and 10% random permutation perform the best as they resemble the real data well but lack privacy protection for real world use. On the other end, 90% random permutation is likely to have higher privacy but can lead to unusable data in datasets with multitude of values for each column. Synthetic data generators aimed at preserving privacy in medical datasets such as HealthGAN perform relatively worse than the baseline models but do ensure privacy protection along with overall utility. However, the skewness affects the observed results in the synthetic data and less prevalent data struggles to be reproduced by synthetic data generation. These results underscore that synthetic datasets might not be

effective for all sub-groups and must be evaluated for biases and unfairness before dissemination. One solution is the development of representative fairness metrics using such time-series metrics that can measure unfairness [24]. This, however, requires a deeper analysis on synthetic data fairness which is not the scope of this paper and is left as future work.

The results presented in this paper highlight the effectiveness of time-series metrics in capturing real and synthetic resemblance. As a future work, the metrics can be used during synthetic time-series data generation to generate higher resemblance datasets, either through hyper-parameter selection via validation, or possibly to construct new loss functions specifically for use to generate time-series data. The research can also be extended to other metrics based on the data composition and can guide future time-series data generation research.

CRediT authorship contribution statement

Karan Bhanot: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Joseph Pedersen:** Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Isabelle Guyon:** Conceptualization, Methodology, Project administration, Supervision, Validation, Writing - review & editing. **Kristin P. Bennett:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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